

COMPLETE COURSE COMPANION · DENNIS G. ZILL (7TH ED.) VS. DR. A. HAGHIGHAT LECTURES

Chapter 2: First-Order Differential Equations

A unified, visual, and rigorous master guide covering qualitative direction fields, autonomous stability, analytical solution protocols (Separable, Linear, Exact, Substitutions), and real-world engineering modeling.

Zill Chapter 2 (pp. 65–124)

Dr. Haghighat Lectures 2, 3, 4

Midterm 1 High-Yield Core

KaTeX Rendered

1 Section 2.1: Direction Fields & Qualitative Analysis

PLAIN-ENGLISH INTUITION

You don't need to integrate a differential equation to see where every solution curve travels! The differential equation $\frac{dy}{dx} = f(x, y)$ serves as a local **slope compass**. At any coordinate point (x, y) , evaluating $f(x, y)$ dictates the exact tangent tilt of a tiny line segment (lineal element). Following the slope lines reveals the complete family of solution curves without doing a single integral.

FORMAL DEFINITION · DIRECTION FIELD & LINEAL ELEMENTS

For a first-order ODE $\frac{dy}{dx} = f(x, y)$, a **direction field** (or slope field) is a grid of points (x_i, y_i) where each point has a short line segment with slope $m = f(x_i, y_i)$.

- A curve $y = \phi(x)$ that solves the DE is everywhere tangent to these lineal elements.
- Uniqueness Geometry:** Two solution curves can never cross or intersect at any point where Picard's Theorem holds.

ISOCINES · RAPID MANUAL SKETCHING

An **isocline** is a curve in the xy -plane along which all tangent slopes are identical constant values c :

$$f(x, y) = c \quad (c = \text{constant slope})$$

- Method:** Choose several constants $c \in \{-2, -1, 0, 1, 2\}$, graph the curves $f(x, y) = c$, and draw parallel tick marks of slope c across each curve.
- Nullclines** ($c = 0$): Where $f(x, y) = 0$, all tangent segments are purely horizontal!

ENGINEERING ANALOGY · WIND & OCEAN CURRENT NAVIGATION

Think of a direction field like a nautical wind or ocean current chart. If a boat cuts its engine and floats freely, the local water current vectors determine its path at every second. The differential equation is simply the formula giving current velocity and angle at every latitude and longitude!

Worked Example · Direction Field by Isoclines (Lecture 2, Slide 11)

CONCORDIA SLIDE EX

Problem: Analyze the direction field of $\frac{dy}{dx} = 0.2xy$ using the method of isoclines.

1 Set up Isocline Equation: $0.2xy = c \implies xy = 5c$.

- For $c = 0$: $xy = 0 \implies$ The coordinate axes ($x = 0$ or $y = 0$) are nullclines with horizontal slopes ($m = 0$).
- For $c = 1$: $xy = 5 \implies$ Rectangular hyperbola $y = 5/x$ with slope $m = +1$.
- For $c = -1$: $xy = -5 \implies$ Hyperbola in Q2 and Q4 with slope $m = -1$.

2 Exact Analytical Comparison:

Separating variables: $\frac{dy}{y} = 0.2x \, dx \implies \ln|y| = 0.1x^2 + C_1 \implies y(x) = Ce^{0.1x^2}$.

Notice how the analytical solutions are symmetric bell-shaped curves expanding outward, aligning 100% with the hyperbolic isocline slopes!

2 Section 2.1 (Cont.): Autonomous Differential Equations & Stability

PLAIN-ENGLISH INTUITION

An equation is **autonomous** ("self-governing") when the rate of change depends *only* on the state y , completely independent of time or position x : $\frac{dy}{dx} = f(y)$. Because slope does not change with x , all lineal elements along any horizontal line are identical. We can compress the entire 2D behavior into a single 1D **phase line** and predict long-term equilibrium with zero integration!

CRITICAL POINTS & EQUILIBRIUM SOLUTIONS

A real zero c of the function $f(y)$ such that $f(c) = 0$ is called a **critical point** (or equilibrium / stationary point).

- The constant horizontal line $y(x) \equiv c$ is an exact **equilibrium solution**.
- If an initial state starts exactly at $y(0) = c$, it remains at $y = c$ for all time ($dy/dx = 0$).

1D PHASE LINE PROTOCOL

- Solve $f(y) = 0$ to find all critical points $\{c_1, c_2, \dots\}$. Plot them as solid dots on a vertical axis.
- These points divide the y -axis into open intervals.
- Pick a test point in each interval and evaluate the sign of $f(y)$:
 - $f(y) > 0 \implies \frac{dy}{dx} > 0 \implies y(x)$ increases (draw $\uparrow$).
 - $f(y) < 0 \implies \frac{dy}{dx} < 0 \implies y(x)$ decreases (draw $\downarrow$).

The Three Stability Classifications (Lecture 2, Slide 17 & Zill p. 69)

Classification	Phase Line Arrows	Physical Behavior as $x \rightarrow \infty$	Visual Geometry
Attractor (Asymptotically Stable)	$\downarrow \bullet \uparrow$ (pointing toward c)	Solutions on both sides approach $y = c$ as $x \rightarrow \infty$. Highly resilient to disturbances.	Horizontal funneling; solutions sandwich toward the line $y = c$.
Repeller (Unstable)	$\uparrow \bullet \downarrow$ (pointing away from c)	Solutions diverge away from $y = c$ as $x \rightarrow \infty$. Any perturbation drives it far away.	Solutions flare away vertically toward $\pm\infty$ or other attractors.
Semi-Stable	$\uparrow \bullet \uparrow$ or $\downarrow \bullet \downarrow$ (same direction)	Solutions approach c from one side, but diverge away on the opposite side.	One-sided attractor; precarious equilibrium.

Comprehensive Worked Example • Multi-Point Stability Analysis

EXAM FAVORITE

Problem: Find the critical points, construct the 1D phase line, and classify the stability of all equilibrium solutions for $\frac{dy}{dx} = y(y-2)^2(4-y)$.

1 Find Critical Points:

$$f(y) = y(y-2)^2(4-y) = 0 \implies y = 0, \quad y = 2, \quad y = 4.$$

2 Sign Analysis across Intervals:

- $y > 4$: Test $y = 5 \implies (5)(3)^2(-1) = -45 < 0 \implies \downarrow$
- $2 < y < 4$: Test $y = 3 \implies (3)(1)^2(+1) = +3 > 0 \implies \uparrow$
- $0 < y < 2$: Test $y = 1 \implies (1)(-1)^2(+3) = +3 > 0 \implies \uparrow$
- $y < 0$: Test $y = -1 \implies (-1)(-3)^2(+5) = -45 < 0 \implies \downarrow$

3 Phase Line & Classification:

- At $y = 4$:** Above $\downarrow$, Below $\uparrow$. Arrows point inward $\implies$ **Asymptotically Stable (Attractor)**.
- At $y = 2$:** Above $\uparrow$, Below $\uparrow$. Arrows point upward on both sides $\implies$ **Semi-Stable**.
- At $y = 0$:** Above $\uparrow$, Below $\downarrow$. Arrows point outward away from 0 $\implies$ **Unstable (Repeller)**.

Long-Term Prediction: If $y(0) = 1.5$, $\lim_{x \rightarrow \infty} y(x) = 4$ because y crosses through the semi-stable threshold at $y = 2$ and gets captured by $y = 4$!

DEADLY EXAM TRAP • MULTIPLICITY RULE FOR STABILITY

If a critical point c comes from a root $(y - c)^k$: If k is **odd**, the sign of $f(y)$ changes across c (yielding Attractor or Repeller). If k is **even**, $(y - c)^k \geq 0$ never changes sign, guaranteeing that c is **Semi-Stable**! In the example above, $(y - 2)^2$ has power 2, immediately proving $y = 2$ is semi-stable without calculations!

3 Section 2.2: Separable Equations & Lost Singular Solutions

PLAIN-ENGLISH INTUITION

Separable equations are the simplest analytical ODEs: you quarantine all terms containing y on the left with dy , and all terms containing x on the right with dx . Once separated, both sides are integrated independently. However, **algebraic division destroys constant solutions!** Whenever you divide by $h(y)$, you must independently test $h(y) = 0$ to avoid losing valid equilibrium solutions.

STANDARD FORM & DEFINITION

A first-order DE is **separable** if the derivative factors into a function purely of x multiplied by a function purely of y :

$$\frac{dy}{dx} = g(x)h(y) \iff p(y)\frac{dy}{dx} = g(x)$$

where $p(y) = \frac{1}{h(y)}$. In differential notation: $p(y) dy = g(x) dx$.

RIGOROUS 4-STEP SEPARATION PROTOCOL

- 1. Roots Check:** Find roots $h(y) = 0$. Each root $y = r$ is a valid constant solution!
- 2. Separation:** Divide by $h(y)$ and multiply by dx : $\frac{dy}{h(y)} = g(x) dx$.
- 3. Integration:** Integrate both sides: $\int \frac{dy}{h(y)} = \int g(x) dx + C$.
- 4. Solve Explicitly:** Isolate y if possible, and check if constant solutions can be recovered by choosing a specific value for C .

THE DEADLY TRAP · THE LOST SINGULAR SOLUTION

If a constant solution $y = r$ (from $h(r) = 0$) cannot be obtained by setting C to any finite real number in your general family $y = \phi(x, C)$, it is a **singular solution**. Failing to state $y = r$ as a separate solution will cause you to lose points on exams!

Master Worked Example · Partial Fractions & Lost Solutions (Lecture 3, Slide 6)

MIDTERM FAVORITE

Problem: Solve the initial value problem $\frac{dy}{dx} = y^2 - 4$, with initial condition $y(0) = 0$. State any singular solutions.

1 Find Constant Solutions (Before Dividing!):

$$y^2 - 4 = 0 \implies (y - 2)(y + 2) = 0 \implies y = 2 \text{ and } y = -2.$$

2 Separate Variables:

$$\frac{dy}{(y-2)(y+2)} = dx.$$

3 Partial Fractions Decomposition:

$$\frac{1}{(y-2)(y+2)} = \frac{A}{y-2} + \frac{B}{y+2} \implies 1 = A(y+2) + B(y-2).$$

$$\text{Setting } y = 2 \implies A = 1/4. \text{ Setting } y = -2 \implies B = -1/4.$$

$$\frac{1}{4} \int \left(\frac{1}{y-2} - \frac{1}{y+2} \right) dy = \int dx$$

4 Integrate & Apply Log Rules:

$$\frac{1}{4} \ln \left| \frac{y-2}{y+2} \right| = x + c_1 \implies \ln \left| \frac{y-2}{y+2} \right| = 4x + 4c_1.$$

$$\text{Exponentiate: } \left| \frac{y-2}{y+2} \right| = e^{4c_1} e^{4x} \implies \frac{y-2}{y+2} = C e^{4x} \text{ (where } C = \pm e^{4c_1}).$$

5 Isolate Explicit $y(x)$:

$$y - 2 = C e^{4x} (y + 2) \implies y(1 - C e^{4x}) = 2(1 + C e^{4x}).$$

$$y(x) = 2 \frac{1 + C e^{4x}}{1 - C e^{4x}}$$

6 Apply IVP & Inspect Singular Solutions:

$$y(0) = 0 \implies 2 \frac{1+C}{1-C} = 0 \implies C = -1.$$

$$\text{Particular Solution: } y(x) = 2 \frac{1 - e^{4x}}{1 + e^{4x}} = -2 \tanh(2x).$$

Singular Solutions Check: $C = 0$ gives $y = 2$. But $y = -2$ requires $C \rightarrow \infty$, so $y = -2$ is a **singular solution!**

4 Section 2.3: Linear Equations & Integrating Factor Protocol

PLAIN-ENGLISH INTUITION

A first-order equation is **linear** if y and y' appear purely to the first power and are never multiplied together. Even when terms cannot be separated, we can exploit the Product Rule in reverse! By multiplying through by a tailor-made **integrating factor** $\mu(x) = e^{\int P(x)dx}$, the entire left side automatically collapses into the single derivative $\frac{d}{dx}[\mu(x)y]$. One integral then solves the entire equation!

STEP 0: STANDARD CANONICAL FORM (MANDATORY)

Before doing anything, divide every term by the leading coefficient $a_1(x)$ to ensure the coefficient of $\frac{dy}{dx}$ is exactly 1:

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \xrightarrow{\div a_1(x)} \frac{dy}{dx} + P(x)y = f(x)$$

where $P(x) = \frac{a_0(x)}{a_1(x)}$ and $f(x) = \frac{g(x)}{a_1(x)}$.

DERIVATION OF INTEGRATING FACTOR $M(X)$

We require $\mu(x) [y' + P(x)y] = \frac{d}{dx}[\mu(x)y] = \mu y' + \mu' y$.

Matching coefficients of y : $\mu' = \mu P(x) \implies \frac{d\mu}{\mu} = P(x)dx$.

$$\ln |\mu| = \int P(x) dx \implies \mu(x) = e^{\int P(x) dx}$$

Note: No $+C$ is needed here; any non-zero integrating factor works!

General Solution Formula & Structure

$$y(x) = \frac{1}{\mu(x)} \left[\int \mu(x)f(x) dx + C \right] = \underbrace{C e^{-\int P(x)dx}}_{y_h(x) \text{ (homogeneous)}} + \underbrace{\frac{1}{\mu(x)} \int \mu(x)f(x) dx}_{y_p(x) \text{ (particular)}}$$

Transient Terms: If $\lim_{x \rightarrow \infty} y_h(x) = 0$ (e.g., negative exponentials e^{-kx}), the homogeneous response dies out, leaving only the steady-state response $y_p(x)$.

Master Worked Example • Standard Form & IVP (Lecture 3, Slide 13 & Zill)

STEP-BY-STEP BREAKDOWN

Problem: Solve the IVP: $x \frac{dy}{dx} - 4y = x^5 e^x$, with $y(1) = 0$ for $x > 0$.

1 Put into Standard Form:

Divide by x : $\frac{dy}{dx} - \frac{4}{x}y = x^4 e^x$.

Identify: $P(x) = -\frac{4}{x}$ and $f(x) = x^4 e^x$.

2 Calculate Integrating Factor $\mu(x)$ (Watch the sign!):

$$\int P(x)dx = \int -\frac{4}{x}dx = -4 \ln x = \ln(x^{-4}).$$

$$\mu(x) = e^{\ln(x^{-4})} = x^{-4} = \frac{1}{x^4}$$

3 Multiply Standard Form & Collapse LHS:

$$\frac{1}{x^4} \frac{dy}{dx} - \frac{4}{x^5}y = \frac{1}{x^4}(x^4 e^x) \implies \frac{d}{dx} [x^{-4}y] = e^x.$$

4 Integrate Both Sides (Include $+C$ immediately!):

$$x^{-4}y = \int e^x dx + C = e^x + C \implies y(x) = x^4 e^x + Cx^4.$$

5 Apply Initial Condition $y(1) = 0$:

$$0 = (1)^4 e^1 + C(1)^4 \implies 0 = e + C \implies C = -e.$$

Final Solution: $y(x) = x^4 e^x - ex^4 = x^4(e^x - e)$ on $(0, \infty)$.

THE TOP 2 EXAM BLUNDERS ON LINEAR EQUATIONS

- Calculating $\mu(x)$ before dividing by $a_1(x)$!** If you take $P(x) = -4$ instead of $-4/x$, $\mu(x) = e^{-4x}$, which ruins the entire problem.
- Adding $+C$ at the very end:** Writing $y = \frac{1}{\mu} \int \mu f dx \rightarrow y = x^4 e^x + C$ is WRONG. The constant must be multiplied by x^4 !

5 Section 2.4: Exact Equations & The 5-Step Reconstruction Protocol

PLAIN-ENGLISH INTUITION

An equation $M(x, y)dx + N(x, y)dy = 0$ is **exact** if it is the total differential $df = 0$ of some hidden 2D surface $f(x, y) = C$. Because mixed second partial derivatives of smooth surfaces are always equal ($\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$), we have a foolproof litmus test: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. If this matches, we reconstruct $f(x, y)$ through systematic partial integration!

THEOREM 2.4.1 · CRITERION FOR AN EXACT DIFFERENTIAL

Let $M(x, y)$ and $N(x, y)$ be continuous with continuous first partial derivatives in a rectangular region R . The differential expression $M(x, y)dx + N(x, y)dy$ is **exact** if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

When exact, there exists a potential function $f(x, y)$ such that $\frac{\partial f}{\partial x} = M$ and $\frac{\partial f}{\partial y} = N$. The general solution is $f(x, y) = C$.

The 5-Step Reconstruction Protocol

- 1 Test Exactness:** Check $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.
- 2 Partial Integrate M :** $f(x, y) = \int M(x, y) dx + g(y)$, where $g(y)$ is the unknown integration constant w.r.t. x .
- 3 Differentiate w.r.t. y :** $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} [\int M dx] + g'(y)$.
- 4 Equate to $N(x, y)$ & Solve for $g'(y)$:** All x terms MUST cancel out completely! Integrate $g'(y)dy$ to get $g(y)$.
- 5 State Implicit Solution:** Write $f(x, y) = C$.

Master Worked Example · 5-Step Exact Reconstruction (Lecture 4, Slide 8)

CLASSIC CONCORDIA EXAM

Problem: Solve the initial value problem: $(2xy)dx + (x^2 - 1)dy = 0$, with initial condition $y(0) = 3$.

1 Identify & Test Mixed Partials:

$$M(x, y) = 2xy \implies \frac{\partial M}{\partial y} = 2x.$$

$$N(x, y) = x^2 - 1 \implies \frac{\partial N}{\partial x} = 2x.$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 2x$, the equation is **EXACT**!

2 Integrate M with respect to x (treating y as a constant):

$$f(x, y) = \int 2xy dx + g(y) = x^2 y + g(y)$$

3 Differentiate $f(x, y)$ with respect to y :

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} [x^2 y + g(y)] = x^2 + g'(y).$$

4 Equate to $N(x, y)$ and solve for $g'(y)$:

$$x^2 + g'(y) = x^2 - 1 \implies g'(y) = -1 \text{ (} x^2 \text{ cancelled!)}.$$

Integrate: $g(y) = \int (-1) dy = -y$.

5 Assemble $f(x, y) = C$ & Apply $y(0) = 3$:

$$x^2 y - y = C \implies y(x^2 - 1) = C.$$

$$3(0^2 - 1) = C \implies C = -3.$$

$$\text{Explicit Solution: } y(x) = \frac{-3}{x^2 - 1} = \frac{3}{1 - x^2}.$$

THE GOLDEN SANITY CHECK · THE CANCELLATION LITMUS TEST

In Step 4, when you set $\frac{\partial f}{\partial y} = N(x, y)$, **every single term containing x must vanish!** If you are left with an expression for $g'(y)$ that still has an x inside (e.g., $g'(y) = 2x + y$), STOP immediately. You either made an error in your partial derivatives in Step 1, or integrated incorrectly in Step 2!

6 Section 2.4 (Cont.): Rescuing Non-Exact Equations with Integrating Factors

PLAIN-ENGLISH INTUITION

What happens when $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$? All is not lost! We can often multiply the entire equation by a special "rescue multiplier" μ that balances the mixed partial derivatives and transforms the stubborn equation into a pristine exact differential. The key is testing whether the discrepancy simplifies into a function of x alone or y alone.

CASE 1: INTEGRATING FACTOR $M(X)$ (FUNCTION OF X ALONE)

If the expression $\frac{M_y - N_x}{N}$ simplifies to depend *only* on x :

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N(x, y)} = p(x) \implies \mu(x) = e^{\int p(x) dx}$$

Mnemonic: Division by N gives a function of x . Numerator starts with M_y .

CASE 2: INTEGRATING FACTOR $M(Y)$ (FUNCTION OF Y ALONE)

If the expression $\frac{N_x - M_y}{M}$ simplifies to depend *only* on y :

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M(x, y)} = q(y) \implies \mu(y) = e^{\int q(y) dy}$$

Mnemonic: Division by M gives a function of y . Numerator reversed ($N_x - M_y$).

Master Worked Example · Non-Exact Made Exact (Lecture 4, Slide 10–11)

DR. HAGHIGHAT SLIDE EX

Problem: Solve the non-exact differential equation: $xy \, dx + (2x^2 + 3y^2 - 20) \, dy = 0$.

1 Test Exactness:

$$M = xy \implies M_y = x.$$

$$N = 2x^2 + 3y^2 - 20 \implies N_x = 4x.$$

$$M_y \neq N_x \ (x \neq 4x) \implies \text{NOT EXACT! Discrepancy: } M_y - N_x = -3x.$$

2 Find Integrating Factor:

- Test Case 1: $\frac{M_y - N_x}{N} = \frac{-3x}{2x^2 + 3y^2 - 20}$ (Contains both $x, y \implies$ FAILS!).
- Test Case 2: $\frac{N_x - M_y}{M} = \frac{4x - x}{xy} = \frac{3x}{xy} = \frac{3}{y}$ (Function of y alone!).

$$\mu(y) = e^{\int \frac{3}{y} dy} = e^{3 \ln |y|} = y^3$$

3 Multiply DE by $\mu(y) = y^3$:

$$(xy^4) \, dx + (2x^2y^3 + 3y^5 - 20y^3) \, dy = 0.$$

$$\text{Verify: } M_y = 4xy^3, N_x = 4xy^3. \text{ NOW EXACT!}$$

4 Integrate & Reconstruct $f(x, y)$:

$$f(x, y) = \int xy^4 \, dx + g(y) = \frac{1}{2}x^2y^4 + g(y).$$

$$\frac{\partial f}{\partial y} = 2x^2y^3 + g'(y) = 2x^2y^3 + 3y^5 - 20y^3.$$

$$g'(y) = 3y^5 - 20y^3 \implies g(y) = \frac{1}{2}y^6 - 5y^4.$$

5 Final Implicit Solution:

$$\frac{1}{2}x^2y^4 + \frac{1}{2}y^6 - 5y^4 = C \iff x^2y^4 + y^6 - 10y^4 = C^*$$

THE DEADLY SIGN INVERSION PITFALL

Notice the subtle sign flip between the two formulas! For $\mu(x)$, the numerator is $(M_y - N_x)$. But for $\mu(y)$, the numerator is $(N_x - M_y) = -(M_y - N_x)$. If you forget this inversion and write $\frac{M_y - N_x}{M} = -\frac{3}{y}$, you will obtain $\mu(y) = y^{-3}$, which will completely fail to make the equation exact!

7 Section 2.5: Solutions by Substitution (Bernoulli & Homogeneous)

PLAIN-ENGLISH INTUITION

When an equation is nonlinear and resists direct separation or integrating factors, a clever change of variables can transform it into an equation we already know how to solve. There are three classic substitution archetypes in ENGR 213: **Homogeneous** (ratio y/x), **Bernoulli** (power y^n), and **Linear Arguments** ($Ax + By + C$).

1. Homogeneous Equations

A DE $M dx + N dy = 0$ is homogeneous if M and N have the same algebraic degree:

$$f(tx, ty) = t^k f(x, y).$$

Substitution:

$$y = ux \implies dy = u dx + x du$$

Always reduces to a **separable equation** in u and x !

2. Bernoulli Equations

Nonlinear ODE with an extra y^n factor:

$$\frac{dy}{dx} + P(x)y = f(x)y^n$$

Substitution:

$$u = y^{1-n} \implies \frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx}$$

Always transforms into a **linear ODE** for $u(x)$!

3. Linear Arguments

The derivative depends purely on a linear composite $Ax + By + C$:

$$\frac{dy}{dx} = f(Ax + By + C)$$

Substitution:

$$u = Ax + By + C \implies \frac{du}{dx} = A + B \frac{dy}{dx}$$

Always reduces to a **separable ODE** in u and x !

Master Worked Example · Bernoulli Equation (Zill p. 96 & Course Syllabus)

BERNOULLI MASTERY

Problem: Solve the nonlinear differential equation: $x \frac{dy}{dx} + y = x^2 y^2$ for $x > 0$.

1 Standard Bernoulli Form:

Divide by x : $\frac{dy}{dx} + \frac{1}{x}y = xy^2 \implies P(x) = \frac{1}{x}, \quad f(x) = x, \quad n = 2.$

2 Apply Bernoulli Substitution:

Let $u = y^{1-n} = y^{1-2} = y^{-1} = \frac{1}{y}$.

Differentiating: $\frac{du}{dx} = -y^{-2} \frac{dy}{dx} \implies \frac{dy}{dx} = -y^2 \frac{du}{dx}.$

3 Substitute into Standard Form:

Divide the original DE by y^2 : $y^{-2} \frac{dy}{dx} + \frac{1}{x} y^{-1} = x.$

Since $\frac{du}{dx} = -y^{-2} \frac{dy}{dx}$, multiply by -1 :

$$\frac{du}{dx} - \frac{1}{x}u = -x \quad (\text{Pure Linear in } u!)$$

4 Solve the Linear Equation for $u(x)$:

$$\mu(x) = e^{\int -\frac{1}{x} dx} = e^{-\ln x} = \frac{1}{x}.$$

$$\frac{d}{dx} \left[\frac{1}{x} u \right] = \frac{1}{x} (-x) = -1.$$

$$\text{Integrate: } \frac{u}{x} = -x + C \implies u(x) = -x^2 + Cx.$$

5 Back-Substitute $u = \frac{1}{y}$:

$$\frac{1}{y} = Cx - x^2 \implies y(x) = \frac{1}{Cx - x^2}$$

Note: The trivial constant solution $y \equiv 0$ was lost when dividing by y^2 in Step 3!

THE UNIVERSAL BERNOULLI TRANSFORMATION SHORTCUT

Whenever you have $y' + P(x)y = f(x)y^n$, the substitution $u = y^{1-n}$ **ALWAYS** directly yields the linear differential equation: $\frac{du}{dx} + (1-n)P(x)u = (1-n)f(x)$. You can jump directly to this linear equation on exams, saving 3 minutes of tedious chain-rule algebra!

8 Sections 2.7 & 2.8: Linear & Nonlinear Engineering Modeling

PLAIN-ENGLISH INTUITION

Engineering differential equations are mathematical translations of fundamental conservation laws: Rate of Accumulation = Input Rate – Output Rate. Whether analyzing chemical brine mixing in environmental tanks, thermal dissipation in heat exchangers, or fluid discharge under gravity, the resulting first-order ODEs are solved using the exact tools mastered in Sections 2.2 through 2.4.

1. Newton's Law of Cooling

The rate of temperature change is proportional to the difference with ambient T_m :

$$\frac{dT}{dt} = k(T - T_m), \quad k < 0$$

Solution: $T(t) = T_m + (T_0 - T_m)e^{kt}$. As $t \rightarrow \infty$, $T(t) \rightarrow T_m$ exponentially.

2. Tank Mixing (Brine / Solute)

Accumulation of solute mass $A(t)$:

$$\frac{dA}{dt} = R_{\text{in}} - R_{\text{out}} = c_{\text{in}}r_{\text{in}} - \left(\frac{A(t)}{V(t)}\right)r_{\text{out}}$$

$V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$. Variable volume yields a **linear ODE!**

3. Torricelli's Law (Draining)

Water height $h(t)$ in a tank with hole area a :

$$A_h \frac{dh}{dt} = -ac\sqrt{2gh}$$

Nonlinear power $h^{1/2}$. Always solved by **separation of variables!**

Master Worked Example • Variable-Volume Mixing Tank (Zill p. 106 & Midterm)

HIGH-YIELD APPLICATION

Problem: A large 300-gallon tank initially contains 100 gallons of pure water ($A(0) = 0$). Brine containing 2 lbs of salt per gallon flows in at $r_{\text{in}} = 3$ gal/min. The well-stirred mixture pumps out at $r_{\text{out}} = 2$ gal/min. Find the amount of salt $A(t)$ at any time, and determine the exact salt content when the tank is completely full.

1 Volume as a Function of Time:

$$V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t = 100 + (3 - 2)t = 100 + t \text{ gallons}$$

The tank overflows when $V(t) = 300 \Rightarrow 100 + t = 300 \Rightarrow t = 200$ min.

2 Formulate the Differential Equation:

$$R_{\text{in}} = c_{\text{in}} \cdot r_{\text{in}} = (2 \text{ lb/gal})(3 \text{ gal/min}) = 6 \text{ lb/min.}$$

$$R_{\text{out}} = c_{\text{out}} \cdot r_{\text{out}} = \left(\frac{A(t)}{100+t}\right)(2) = \frac{2}{100+t}A(t).$$

$$\frac{dA}{dt} + \frac{2}{100+t}A = 6 \quad (\text{Standard Linear ODE!})$$

3 Solve via Integrating Factor:

$$\mu(t) = e^{\int \frac{2}{100+t} dt} = e^{2 \ln(100+t)} = (100+t)^2.$$

$$\frac{d}{dt} [A(100+t)^2] = 6(100+t)^2.$$

$$\text{Integrate: } A(100+t)^2 = 6 \frac{(100+t)^3}{3} + C = 2(100+t)^3 + C.$$

$$A(t) = 2(100+t) + \frac{C}{(100+t)^2}$$

4 Apply Initial Condition $A(0) = 0$:

$$0 = 2(100) + \frac{C}{100^2} \Rightarrow C = -200 \times 10,000 = -2,000,000.$$

$$\text{Salt Equation: } A(t) = 2(100+t) - \frac{2 \times 10^6}{(100+t)^2} \text{ for } t \in [0, 200].$$

At Overflow ($t = 200$ min):

$$A(200) = 2(300) - \frac{2,000,000}{300^2} = 600 - 22.22 = 577.8 \text{ lbs.}$$

THE CLASSIC MIXING TANK BLUNDER

When inflow rate $\neq$ outflow rate ($r_{\text{in}} \neq r_{\text{out}}$), the tank volume is dynamic! Students who mistakenly write $V = 100$ and use $\frac{dA}{dt} + \frac{2}{100}A = 6$ lose full marks. Always write $V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$ first!

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Synthesis: Master 10-Second Diagnostic Tree & The 7 Deadly Exam Traps

EXAM TAKING STRATEGY

On an ENGR 213 exam, 70% of errors occur because students choose the wrong solution technique or fail to normalize equations into standard form. Use this 10-second hierarchical filter before putting pen to paper!

The 10-Second First-Order Differential Equation Diagnostic Filter			
Priority	Equation Archetype	How to Spot It	Standard Transformation / Method
1. Separable (Section 2.2)	$\frac{dy}{dx} = g(x)h(y)$	x 's and y 's are factored into clean products/quotients.	$\int \frac{dy}{h(y)} = \int g(x)dx$. Always check $h(y) = 0$ for lost singular solutions!
2. Linear (Section 2.3)	$a_1(x)y' + a_0(x)y = g(x)$	y and y' appear purely to 1st power, never inside nonlinear functions.	Divide by $a_1(x) \rightarrow y' + Py = f$. Integrating factor $\mu(x) = e^{\int P(x)dx}$.
3. Exact (Section 2.4)	$M(x,y)dx + N(x,y)dy = 0$	Sum of differentials equals zero.	Test $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. If equal, reconstruct $f(x,y) = \int Mdx + g(y) = C$.
4. Non-Exact (Section 2.4)	$Mdx + Ndy = 0$, $M_y \neq N_x$	Exactness test fails, but differences factor cleanly.	$\frac{M_y - N_x}{N} = p(x) \implies \mu(x)$ OR $\frac{N_x - M_y}{M} = q(y) \implies \mu(y)$.
5. Bernoulli (Section 2.5)	$y' + P(x)y = f(x)y^n$	Looks linear except for the pesky y^n on the RHS.	Substitute $u = y^{1-n} \implies u' + (1-n)Pu = (1-n)f$ (Linear!).
6. Homogeneous (Section 2.5)	$M(tx,ty) = t^k M(x,y)$	All polynomial terms in M and N have identical total degree.	Substitute $y = ux \implies dy = udx + xdu$ (becomes Separable!).
7. Linear Argument (Section 2.5)	$y' = f(Ax + By + C)$	Arguments of trig/exp are linear combinations $Ax + By + C$.	Substitute $u = Ax + By + C \implies u' = A + By'$ (becomes Separable!).

THE 7 DEADLY EXAM TRAPS & ANTI-MISTAKE CHECKLIST

1. **Forgetting Standard Form:** Calculating $\mu(x) = e^{\int a_0(x)dx}$ without dividing by $a_1(x)$ first in linear equations.

2. **The Vanishing Constant Solution:** Forgetting to list $y = r$ as a singular solution when dividing by $h(y)$ in separable ODEs.

3. **The Multiplier Sign Flip:** Writing $\frac{M_y - N_x}{M}$ instead of $\frac{N_x - M_y}{M}$ when finding $\mu(y)$ for non-exact equations.

4. **Premature +C Neglect:** Writing $y = \frac{1}{\mu} \int \mu f dx + C$ instead of multiplying C by $\frac{1}{\mu(x)}$.

5. **Product Rule Omission:** Forgetting the second term when differentiating $y = ux \implies dy = udx + xdu$ (writing $dy = udx$).

6. **Static Volume in Tank Mixtures:** Assuming constant volume when $r_{in} \neq r_{out}$ instead of $V(t) = V_0 + (r_{in} - r_{out})t$.

7. **Absolute Value Drop in Logarithms:** Dropping absolute values $\ln|y|$ prematurely, causing branch errors for negative initial conditions.

GOLDEN RULE FOR MIDTERM 1 PREPARATION

Every first-order ODE question on the ENGR 213 midterm tests two things: (1) **Pattern Recognition** (identifying which box in the table above fits the problem), and (2) **Meticulous Bookkeeping** (carrying negative signs and integration constants through every algebraic step). Check your candidate solution by plugging it back into the original differential equation whenever time permits!